Scratch Cat Duo

Studio 87. Free and open source (AGPL-3.0-or-later).

Intro

Scratch Cat Duo is two DJ decks and a mixer inside one plugin. Load a track on each deck (it is analysed for BPM, beat grid and key), or send a DAW track to a deck and scratch the last minute of it. Use it to scratch, mix and loop your own songs while you compose, or to play a set from your DAW.

Quick start

- Click the track name box on a deck (or use the track browser at the bottom, or drop a file) and pick a file (wav, aiff, flac, ogg or mp3). The BPM and the key (Camelot code and name) appear next to the deck name.
- Press PLAY. Drag the platter to scratch: the sound follows your hand, forward and backward, and holds when you stop.
- Move the crossfader between the decks. Sync matches the deck tempo to the DAW tempo (or to the other deck when Tempo is set to Decks lead).
- The two waveforms at the top show both decks on one time axis, so when the beat lines match the beats are aligned.

Deck

- Source: File plays the loaded track. DAW input plays the audio that arrives on the plugin input for that deck; the last minute is kept so you can scratch it. Go live returns to the live sound.
- CUE: while playing it jumps back to the cue point and stops. While stopped it sets the cue point here; hold it on the cue point to preview.
- Sync, Key lock and Quant. Key lock changes the tempo without changing the pitch. Quant snaps cue points, hot cues and loops to the beat grid.
- Tempo fader with a range of 8, 16 or 50 %. Up is faster, down is slower. Double click to reset.
- Loop in, Loop out and Exit, and beat loops of 1/4 to 8 beats from the playhead.
- Brake slows to a stop. Backspin spins backward and stops. Roll, Censor, Stutter and Trans work while you hold them; Roll and Censor keep the track running underneath and jump back in time when released.
- Eight hot cue pads: an empty pad sets a cue at the playhead, a set pad jumps there, right click clears it.

Mixer

- Trim, three band EQ (High, Mid, Low; all the way down kills the band), Colour filter (left is a low pass, right is a high pass, centre is off), CUE for the headphone output, and the channel fader.
- Crossfader with three curves: Smooth (linear), Sharp (both decks at full level, then a quick cut near each end) and Const (constant power).
- The master level, and the safety soft clip that only acts above 0.95.

FX chain

- Three effect slots in series: Filter, Echo, Reverb, Flanger or Crush, each with an On switch, an Amount and a Dry/Wet. Apply the chain to Deck A, Deck B, A + B or the master. Chain mix scales every slot. Echo time is set in beats and follows the tempo.

DAW routing and MIDI

- Deck A in and Deck B in are the plugin inputs. Enable the extra inputs and outputs in your host to route DAW tracks in and each deck out to its own channel. Outputs set to A, B and Master gives each deck its own stereo output next to the master.
- Cue output: the channels with CUE on, mixed with the master by the Cue mix knob.

- Tempo: Follow DAW uses the host tempo for Sync and for the echo; Decks lead uses the other deck.
- MIDI learn: pick a control in the list, press Learn, then move or press the control on your controller. Jog wheel mapping is relative; set its sensitivity in the gear menu. Mappings are saved with the project. Built-in maps for specific controllers are planned.

Window

- Resize by dragging any edge or corner; the 100% button returns to the normal size.
- Hover a control to see an info bubble. Turn them off in the gear menu.
- Free and open source. The plugin collects no data and never touches the network.

Interface tour

Every pink number marks a control that has an info bubble in the interactive mockup (docs/mockup), and the tables below carry the text of those bubbles. The pictures come from the mockup, which mirrors the plugin window; the compiled plugin can differ by a few pixels of layout.



Header bar

1	**Previous preset**	Previous preset.
2	**Init**	Open the preset list: factory presets and your own.
3	**Next preset**	Next preset.
4	**Save**	Save the current settings as your own preset (stays on this device).
5	**Delete, export or import your presets**	Delete, export or import your presets.
6	**Guide**	Quick guide: what each part does.
7	**Settings**	Info bubbles and eco mode.
8	**About & Support**	About Studio 87, privacy, and ways to support the project.

Window

9	**Both**	Start or stop both decks together. Shortcut: Space.
10	**Restart both**	Jump both decks to the start so the beats line up again.
11	**4 bars**	Show this many bars across the waveforms.
12	**Show this many bars across the waveforms**	Show this many bars across the waveforms.
13	**16**	Show this many bars across the waveforms.
14	**Analyze key & BPM**	Detect the tempo and key of this deck again (shown as BPM, key and Camelot code).
15	**File**	Source of this deck. File: plays once and stops at the end. DAW: loops what the host sends. Sampler: loops, with the sampler pads ready.
16	**DAW**	Source of this deck. File: plays once and stops at the end. DAW: loops what the host sends. Sampler: loops, with the sampler pads ready.
17	**Sampler**	Source of this deck. File: plays once and stops at the end. DAW: loops what the host sends. Sampler: loops, with the sampler pads ready.
18	**The track on this deck**	The track on this deck. Click to load your own file (wav, mp3, flac, aiff, ogg). It stays in this browser tab.
19	**Sync**	Match this deck's tempo to the host BPM (or to the master deck) and keep it matched.
20	**Master**	Make this deck the tempo master: the other deck's Sync follows it.
21	**Key lock**	Key lock: change the tempo without changing the pitch.
22	**Quant**	Quantize: cue points, hot cues and loops snap to the nearest beat.
23	**Loop of 1/4 beat from here**	Loop of 1/4 beat from here. Press again to leave the loop.
24	**Loop of 1/2 beat from here**	Loop of 1/2 beat from here. Press again to leave the loop.
25	**Loop of 1 beat from here**	Loop of 1 beat from here. Press again to leave the loop.
26	**Loop of 2 beats from here**	Loop of 2 beats from here. Press again to leave the loop.
27	**Loop of 4 beats from here**	Loop of 4 beats from here. Press again to leave the loop.
28	**Loop of 8 beats from here**	Loop of 8 beats from here. Press again to leave the loop.
29	**Loop in**	Set the loop start here.
30	**Loop out**	Set the loop end here and start looping.
31	**Exit**	Leave the loop.
32	**Vinyl**	On: pausing slows the record down like a turntable. Off: it stops at once.
33	**Trans.**	Transform gate: chops the sound on and off in time with the beat.
34	**Stutter**	Hold: retriggers the cue point every 1/16 beat.
35	**Brake**	Slows the record to a stop.
36	**Backspin**	The record spins back fast, then stops.
37	**Roll**	Hold: rolls a tiny loop of 1/8 beat; letting go lands where the song would have been.
38	**Censor**	Hold: plays backwards; letting go jumps to where the song would have been.
39	**CUE**	CUE. While playing: jumps back to the cue point and pauses. While paused: sets the cue point here; hold to preview from it.
40	**Play or pause the deck**	Play or pause the deck.
41	**Tempo**	Speed of this deck. Up is faster. The range buttons set how far it goes. Double-click resets.
42	**How far the tempo fader goes (percent)**	How far the tempo fader goes (percent).

43	**16**	How far the tempo fader goes (percent).
44	**50**	How far the tempo fader goes (percent).
45	**Reset**	Put the tempo fader back to zero.
46	**Hot Cue**	What the eight pads do: hot cues, loops, jumps by beats, sampler hits or momentary effects.
47	**Loop**	What the eight pads do: hot cues, loops, jumps by beats, sampler hits or momentary effects.
48	**Jump**	What the eight pads do: hot cues, loops, jumps by beats, sampler hits or momentary effects.
49	**Sampler**	What the eight pads do: hot cues, loops, jumps by beats, sampler hits or momentary effects.
50	**Pad FX**	What the eight pads do: hot cues, loops, jumps by beats, sampler hits or momentary effects.
51	**Delete**	Delete switch: while on, clicking a set hot cue removes it. Holding Shift while clicking does the same.
52	**Hot cue 1**	Hot cue 1. Empty: click to set a marker here. Set: click to jump to it. Shift-click, or the Delete switch, removes it. Right-click edits name, colour, position and on/off.
53	**Hot cue 2**	Hot cue 2. Empty: click to set a marker here. Set: click to jump to it. Shift-click, or the Delete switch, removes it. Right-click edits name, colour, position and on/off.
54	**Hot cue 3**	Hot cue 3. Empty: click to set a marker here. Set: click to jump to it. Shift-click, or the Delete switch, removes it. Right-click edits name, colour, position and on/off.
55	**Hot cue 4**	Hot cue 4. Empty: click to set a marker here. Set: click to jump to it. Shift-click, or the Delete switch, removes it. Right-click edits name, colour, position and on/off.
56	**Hot cue 5**	Hot cue 5. Empty: click to set a marker here. Set: click to jump to it. Shift-click, or the Delete switch, removes it. Right-click edits name, colour, position and on/off.
57	**Hot cue 6**	Hot cue 6. Empty: click to set a marker here. Set: click to jump to it. Shift-click, or the Delete switch, removes it. Right-click edits name, colour, position and on/off.
58	**Hot cue 7**	Hot cue 7. Empty: click to set a marker here. Set: click to jump to it. Shift-click, or the Delete switch, removes it. Right-click edits name, colour, position and on/off.
59	**Hot cue 8**	Hot cue 8. Empty: click to set a marker here. Set: click to jump to it. Shift-click, or the Delete switch, removes it. Right-click edits name, colour, position and on/off.
61	**Trim**	Input gain of deck A. Double-click resets.
62	**HI**	High shelf (4 kHz). Double-click resets.
63	**MID**	Mid band (1 kHz). Double-click resets.
64	**LOW**	Low shelf (250 Hz). Double-click resets.
65	**COLOR**	One-knob filter: left is low-pass, right is high-pass. Double-click resets.
66	**Kill the highs of deck A**	Kill the highs of deck A.
67	**Kill the mids of deck A**	Kill the mids of deck A.
68	**Kill the lows of deck A**	Kill the lows of deck A.
69	**CUE**	Headphone cue: sends this deck to the plugin's cue output. The mockup has one output, so this only lights up.
70	**Channel fader of deck A**	Channel fader of deck A. Double-click resets.
71	**Trim**	Input gain of deck B. Double-click resets.
72	**Kill the highs of deck B**	Kill the highs of deck B.
73	**Kill the mids of deck B**	Kill the mids of deck B.
74	**Kill the lows of deck B**	Kill the lows of deck B.

75	**Channel fader of deck B**	Channel fader of deck B. Double-click resets.
76	**Crossfader**	Crossfader: A on the left, B on the right. The thumb sits in the middle when both decks are open. Double-click resets.
77	**Smooth**	Smooth: both full in the middle. Sharp: cuts like a battle mixer. Constant: equal power.
78	**Sharp**	Smooth: both full in the middle. Sharp: cuts like a battle mixer. Constant: equal power.
79	**Const**	Smooth: both full in the middle. Sharp: cuts like a battle mixer. Constant: equal power.
80	**Deck A input**	The audio on deck A. Choose Your file to load your own.
81	**Deck B input**	The audio on deck B. Choose Your file to load your own.
82	**Master only**	Master only gives one stereo out. A, B, Master gives three stereo outs for your DAW's mixer (plugin only; the mockup plays the master).
83	**A · B · Master**	Master only gives one stereo out. A, B, Master gives three stereo outs for your DAW's mixer (plugin only; the mockup plays the master).
84	**Follow DAW**	Follow DAW: Sync uses the host tempo. Decks lead: the master deck sets the tempo and the other deck's Sync follows it.
85	**Decks lead**	Follow DAW: Sync uses the host tempo. Decks lead: the master deck sets the tempo and the other deck's Sync follows it.
86	**HOST BPM**	The tempo your DAW runs at. Sync matches a deck to it. Double-click resets.
87	**Controller**	Choose the DJ controller the plugin listens to. Design mockup only.
100	**Window 100 %**	Window size. Drag any edge or corner of the plugin to resize it; the shape stays the same. Double-click here for 100 %.
101	**Resize**	Drag to resize the window. Every edge and corner works, and the shape stays the same.

Mixer

60	**MASTER**	Output level. Double-click resets.

FX chainDeck ADeck BA + BMaster

88	**Deck A**	Which signal goes through the three effects.
89	**Deck B**	Which signal goes through the three effects.
90	**Which signal goes through the three effe**	Which signal goes through the three effects.
91	**Master**	Which signal goes through the three effects.
92	**What this slot does**	What this slot does.
93	**Turn this effect slot on or off**	Turn this effect slot on or off.
94	**Amount**	Filter: sweep. Echo: delay length. Reverb: size. Flanger: depth. Crush: amount. Double-click resets.
95	**DRY/WET**	How much of the effect is mixed in. Double-click resets.
96	**CHAIN MIX**	Scales the dry/wet of all three effects at once. Double-click resets.

Track browser0 audio files

97	**ADD FOLDER**	Choose a folder. Every subfolder is scanned too. In the plugin this opens the system folder dialog and remembers the path.
98	**ADD FILES**	Add single audio files.
99	**Type to filter the list**	Type to filter the list. Matches the file name, its folder and its format (wav, mp3, flac, aiff...).